

Publication List Leander Thiele

August 1, 2026

- J. Liu, V. Krishnaraj, K. Vovk, K. Aizawa, A.E. Bayer, L. Blot, J. Cowell, S. Garg, J. Gree, A. Hell, B. Horowitz, M. Ichikawa, K. Iemoto, K. Kondo, Z. Lorsin, K. McCarthy, J. Robinson, M. Ruiz-Granda, **L. Thiele**, I. Vovk, M. Zhou, *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology II: Project Planning and Proposal Evaluation*, 2026, [arXiv:2607.25881](#) [cs.CL]
- A. Hell, K. Vovk, V. Krishnaraj, J. Liu, K. Aizawa, A.E. Bayer, L. Blot, J. Cowell, S. Garg, J. Gree, B. Horowitz, M. Ichikawa, K. Iemoto, K. Kondo, Z. Lorsin, K. McCarthy, J. Robinson, M. Ruiz-Granda, **L. Thiele**, I. Vovk, M. Zhou, *AI's Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology I: Literature Review*, 2026, [arXiv:2607.25672](#) [astro-ph.IM]
- B.Y. Wang, **L. Thiele**, M. Ho, *Cluster Mass Inference from Galaxy Kinematics*, 2026, [arXiv:2606.12938](#) [astro-ph.CO]
- L. Thiele**, *Machine Learning Techniques for Astrophysics and Cosmology: Simulation-Based Inference*, 2026, [arXiv:2605.10719](#) [astro-ph.CO], Invited chapter for the edited book "Machine Learning Techniques for Astrophysics and Cosmology" (Eds. C. Bambi, V. Kashyap, S. Shashank, N. Yoshida, Springer Singapore, expected in 2026)
- A. Hell, **L. Thiele**, *LLMs with in-context learning for Algorithmic Theoretical Physics*, 2026, [arXiv:2605.08212](#) [cs.LG], poster at ICML 2026 AI4Physics workshop
- L. Thiele**, *Bayesian Cosmic Void Finding with Graph Flows*, 2026, *OJAp* 9, [arXiv:2602.14630](#) [astro-ph.CO], poster at ICML 2026 AI4Physics workshop
- J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026, *PRD* 114, 023567, [arXiv:2601.20669](#) [astro-ph.CO]
- G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the y -distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025, [arXiv:2512.03038](#) [astro-ph.CO]
- A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025, *JCAP* 07, 023 [arXiv:2511.20423](#) [astro-ph.CO],
- B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025, [arXiv:2510.08573](#) [astro-ph.CO], poster at ML4PS workshop NeurIPS 2025
- B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025, [arXiv:2507.20378](#) [cs.LG], spotlight talk at ML4Astro workshop (colocated with ICML 2025)
- J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025, *MNRAS* 546, 2, [arXiv:2507.12315](#) [astro-ph.CO]

L. Thiele, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025, [arXiv:2507.00514](#) [astro-ph.CO], poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A. LaPosta, et al (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025, JCAP 11, 063, [arXiv:2503.14454](#) [astro-ph.CO]

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024, poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

L. Thiele, *De-baryonifying halos via optimal transport*, 2024, [arXiv:2411.18399](#) [astro-ph.CO]

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A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, D. Chuss, J. Delabrouille, C. Dvorkin, B. Hensley, J.C. Hill, B. Maffei, A. Pullen, A. Rotti, A. Sabyr, **L. Thiele**, E. Wollack, I. Zelko, *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025, JCAP 004, 020, [arXiv:2405.20403](#) [astro-ph.CO]

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D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024, PRD 110, 103539, [arXiv:2403.03807](#) [astro-ph.CO]

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L. Thiele, E. Massara, A. Pisani, C. Hahn, D.N. Spergel, S. Ho, B. Wandelt, *Neutrino mass constraint from an Implicit Likelihood Analysis of BOSS voids*, 2024, ApJ 969, 89, [arXiv:2307.07555](#) [astro-ph.CO]

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L. Thiele, M. Cranmer, W. Coulton, S. Ho, D.N. Spergel, *Predicting the Thermal Sunyaev-Zel'dovich Field using Modular and Equivariant Set-Based Neural Networks*, 2022, MLST 3, 035002, [arXiv:2203.00026 \[astro-ph.CO\]](#), poster at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021)

L. Thiele, D. Wadekar, J.C. Hill, N. Battaglia, J. Chluba, F. Villaescusa-Navarro, L. Hernquist, M. Vogelsberger, D. Anglés-Alcázar, F. Marinacci, *Percent-level constraints on baryonic feedback with spectral distortion measurements*, 2022, PRD 105, 083505, [arXiv:2201.01663 \[astro-ph.CO\]](#)

D. Wadekar, **L. Thiele**, F. Villaescusa-Navarro, J.C. Hill, D.N. Spergel, M. Cranmer, N. Battaglia, D. Anglés-Alcázar, L. Hernquist, S. Ho, *Augmenting astrophysical scaling relations with machine learning: application to reducing the SZ flux-mass scatter*, 2022, PNAS 120(12), [arXiv:2201.01305 \[astro-ph.CO\]](#)

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, L.A. Perez, P. Villanueva-Domingo, D. Wadekar, H. Shao, F.G. Mohammad, S. Hassan, E. Moser, E.T. Lau, L.F.M.P. Valle, A. Nicola, **L. Thiele**, Y. Jo, O.H.E. Philcox, B.D. Oppenheimer, M. Tillman, C. Hahn, N. Kaushal, A. Pisani, M. Gebhardt, A.M. Delgado, J. Caliendo, C. Kreisch, K.W.K. Wong, W.R. Coulton, M. Eickenberg, G. Parimbelli, Y. Ni, U.P. Steinwandel, V. La Torre, R. Dave, N. Battaglia, D. Nagai, D.N. Spergel, L. Hernquist, B. Burkhart, D. Narayanan, B. Wandelt, R.S. Somerville, G.L. Bryan, M. Viel, Y. Li, V. Irsic, K. Kraljic, M. Vogelsberger, *The CAMELS project: public data release*, 2022, ApJS 265, 54, [arXiv:2201.01300 \[astro-ph.CO\]](#)

B. Maffei, M.H. Abitbol, N. Aghanim, J. Aumont, E. Battistelli, J. Chluba, X. Coulon, P. De Bernardis, M. Douspis, J. Grain, S. Gervasoni, J.C. Hill, A. Kogut, S. Masi, T. Matsumara, C. O Sullivan, L. Pagano, G. Pisano, M. Remazeilles, A. Ritacco, A. Rotti, V. Sauvage, G. Savini, S.L. Stever, A. Tartari, **L. Thiele**, N. Trappe, *BISOU: a balloon project to measure the CMB spectral distortions*, 2021, 16th Marcel Grossmann Meeting, [arXiv:2111.00246 \[astro-ph.IM\]](#)

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